Multimodal Media Classifier: Classification of Images, Audio and Text with Python

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Introduction

Every day a vast number of media files are created: photos, recordings and documents. Classifying them, that is, giving each file a label based on its content, is the base of many real systems, like photo galleries that group pictures by color, apps that detect noisy recordings or search engines that find the topic of a document. In this final project we joined the three main topics of the course in one Python program: image processing, audio analysis and text processing. The program reads a folder with 28 real files (10 images, 8 sounds and 10 texts), detects the type of each file automatically and classifies it: the images by color and pattern, the sounds by noise level and dominant pitch, and the texts by word frequencies and keywords.

General objective

- To design a Python program that automatically processes, analyzes and classifies images, audio files and texts based on their attributes.

Specific objectives

- To classify images by their dominant color and pattern using OpenCV.
- To detect the noise level and the dominant pitch of real sounds using Librosa and the Fourier Transform.
- To extract word frequencies, keywords and writing style from texts with NLTK and classify them by topic.
- To show step by step how the color, the edges, the noise level and the pitch are obtained.
- To summarize, compare and visualize the results with Pandas, Matplotlib and Seaborn.

Methods

The data is real and is saved in the Datasets folder: 10 sample images of OpenCV and scikit-image, 8 different recordings of 5 seconds of the ESC-50 dataset of environmental sounds (rain, vacuum cleaner, cow, siren, chirping birds, rooster, car horn and clock alarm) and 10 texts of about 2,000 words of the Brown Corpus of NLTK (4 news, 3 religion and 3 romance texts). The script MediaClassifier_Code.py goes through the folder with os.walk, loads each file only once and sends it to its analysis function depending on its extension. The processing is written in small functions (for example color_steps, edge_steps, fft_spectrum or rms_db), and the same functions are used to classify the files and to draw the step by step figures, so the charts always show what the program really used. The results are saved in Pandas DataFrames, summarized with describe() and groupby(), plotted with Matplotlib and Seaborn, and the class of every file is saved in Results/classification_results.csv.

- Images: each image is read with OpenCV, resized to 512 pixels of width and converted to grayscale and HSV. A hue histogram (cv2.calcHist) of the pixels with a pure color gives the dominant color (red, orange, yellow, green, cyan, blue or purple); if less than 15% of the pixels have a pure color, the image is Grayscale. For the pattern, the image is blurred and the edges are found with the Canny detector, and the percentage of edge pixels gives the class: Smooth (< 3%), Medium texture (3% to 10%) or highly textured (> 10%).

- Audio: each sound is loaded with librosa.load and filtered with a high-pass Butterworth filter of 50 Hz. The RMS energy of each frame is converted to dB: the quietest frames are the background noise, and the loudest ones are the sound, so their difference is the signal-to-noise ratio (SNR). The noise level is High (SNR < 8 dB), Medium (8 to 20 dB) or Low (> 20 dB). The dominant pitch is the frequency with the highest magnitude in the Fourier Transform (fft of SciPy); if it has less than 5% of the energy the sound has no clear pitch, otherwise it is Low (< 500 Hz), Medium (500 to 2,000 Hz) or High (> 2,000 Hz).

- Text: each text is tokenized with word_tokenize and sent_tokenize, converted to lowercase, and the punctuation and English stopwords of NLTK are removed. FreqDist gives the word frequencies, and the program also measures the lexical diversity and the average length of the sentences and words. Each text gets a score for three topics (Politics/News,

Religion and Romance/Fiction): the percentage of its words that are in a list of 16 keywords of that topic, and the topic with the highest score is the class of the text. Finally, the keywords of each topic are the words that are much more frequent in that topic than in the others.

Results

Images

Figure 1 shows the 10 images with their class. By color, 3 images were blue (the baboon, the house with the sky and the Starry Night), 3 grayscale (the coins, the gravel and the moon), 2 orange (the apple and the coffee), 1 green (the circuit board) and 1 red (the smarties). By pattern, 3 were smooth (apple, moon and smarties, with 0.6% of edge pixels on average), 4 had medium texture (5.4%) and 3 were highly textured (circuit board, gravel and Starry Night, 15.0%). This shows that color and pattern are independent: there are smooth and textured images in grayscale and in color.

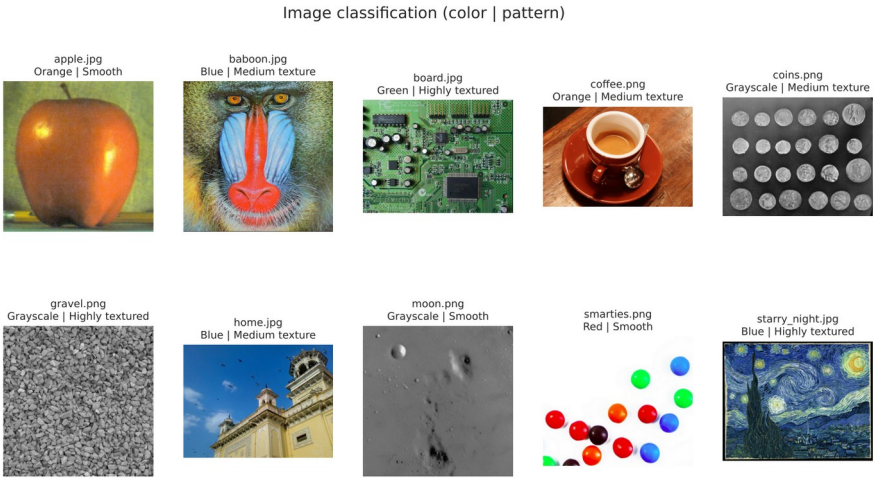


Figure 1. Automatic classification of the images (color | pattern).

Figure 2 shows how the program gets these classes. In the Starry Night, 75% of the pixels have a pure color, and 68% of them are in the blue range, so its dominant color is blue. In the coins, the grayscale image is blurred to remove small noise and Canny finds the borders of the coins, which are 3.6% of the pixels, so the image has medium texture.

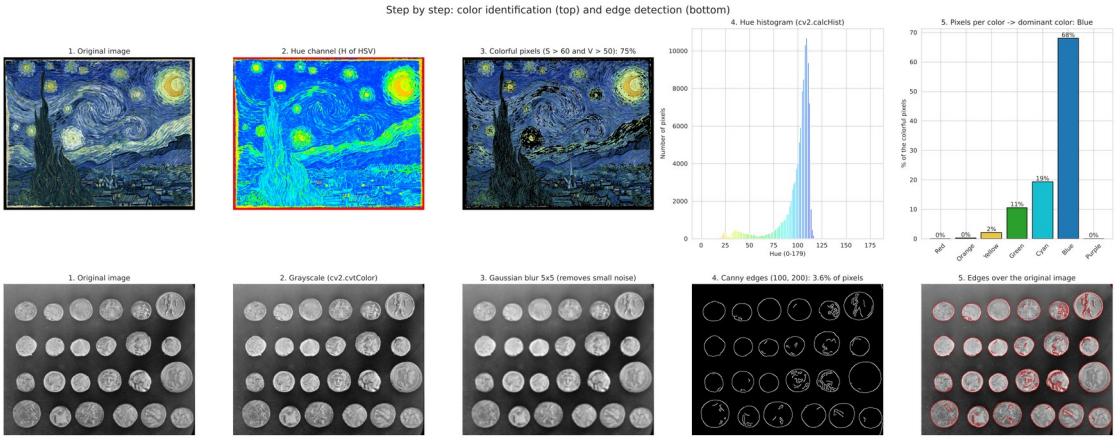


Figure 2. Step by step: color identification (top) and edge detection (bottom).

Audio

Figure 3 shows the process with the cow: the filter removes the rumble under 50 Hz, the background noise is -67.2 dB and the sound is -4.8 dB, so the SNR is 62.4 dB (Low noise); then the FFT finds the dominant pitch at 383 Hz (G4), which has 50.6% of the energy around it, so it is a clear Low pitch. Figure 4 shows the spectrograms of the 8 sounds: the rain and the vacuum cleaner fill all the frequencies all the time, while the other sounds show clear lines, which are their pitch.

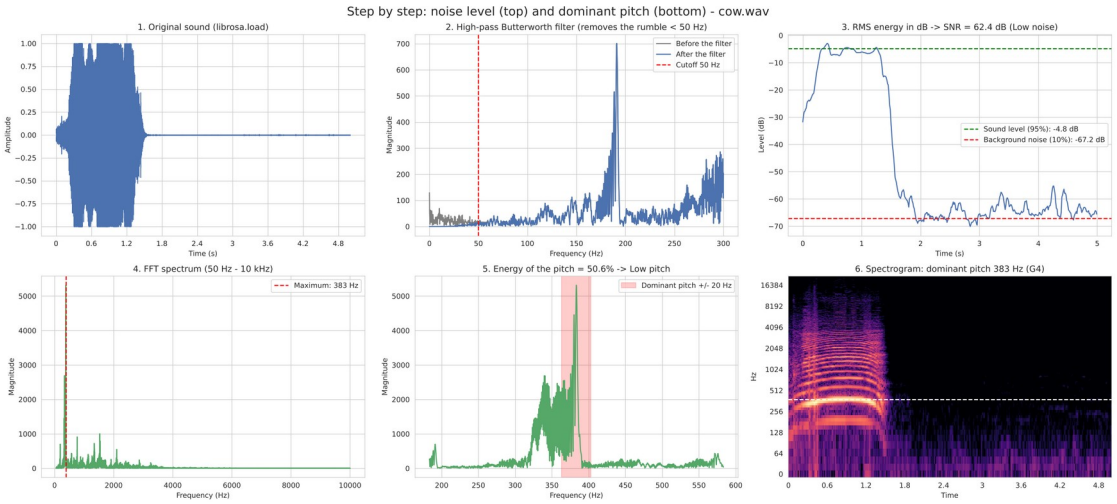


Figure 3. Step by step: noise level (top) and dominant pitch (bottom) of the cow.

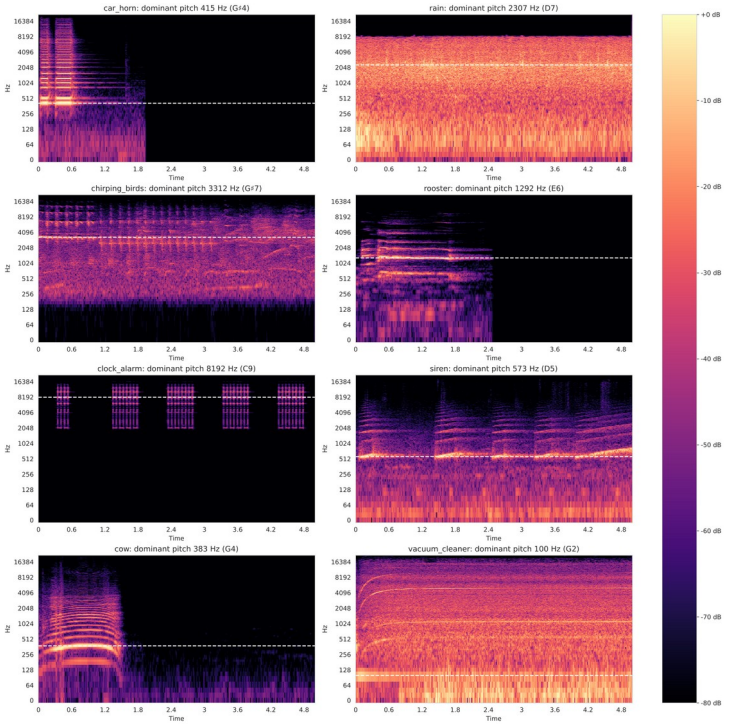


Figure 4. Spectrograms of the 8 sounds; the dashed line is the dominant pitch.

The SNR separated very well the constant noises from the short sounds (Figure 5): the rain (3.1 dB) and the vacuum cleaner (3.4 dB) were High noise and had no clear pitch, the chirping birds were Medium noise (15.6 dB) because the recording has some background, and the other 5 sounds were Low noise, with the car horn, the rooster and the clock alarm above 76 dB because they sound and then there is silence. By pitch, the car horn (415 Hz) and the cow (383 Hz) were Low, the siren (573 Hz) and the rooster (1,292 Hz) were Medium, and the chirping birds (3,312 Hz) and the clock alarm (8,192 Hz, the highest) were High.

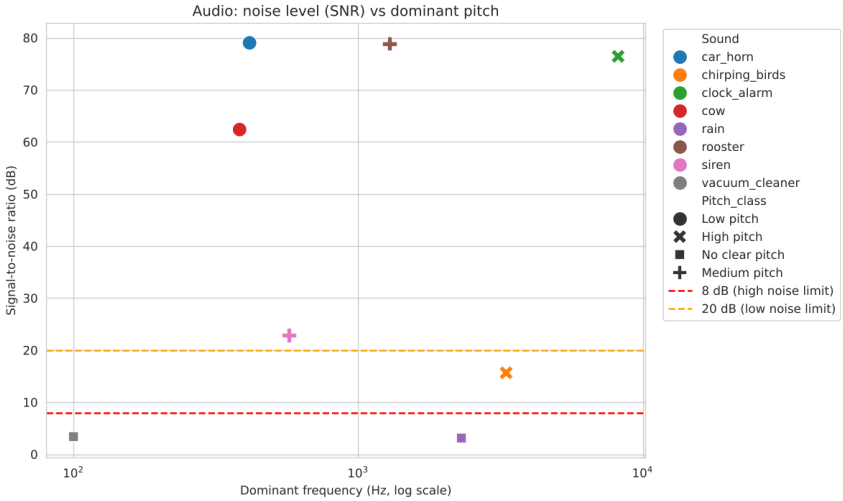


Figure 5. Noise level (SNR) vs dominant pitch of the 8 sounds.

Text

The 10 texts have 19,839 words (about 2,000 each) and the most frequent words, without stopwords, were "would" (117 times), "said" (92) and "one" (63). The keywords of each topic (Figure 6) are very logical: "president", "administration", "jury", "county" and "bill" for Politics/News; "spirit", "catholic", "church" and "community" for Religion; and narrative words and character names like "old", "man", "back", "maggie" and "miss" for Romance/Fiction.

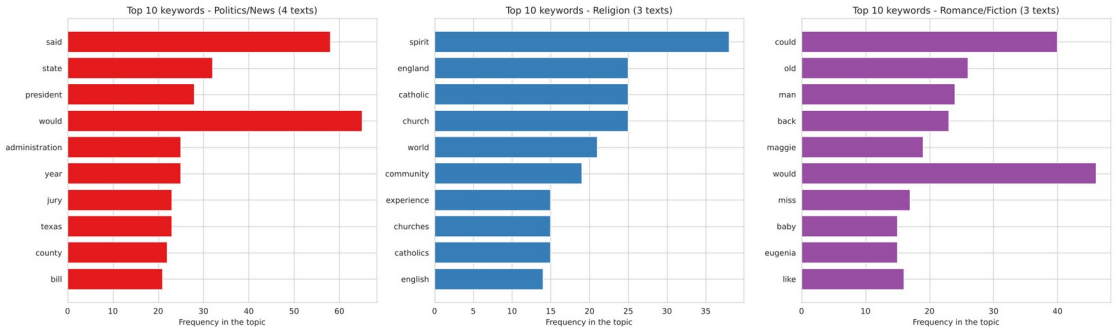


Figure 6. Top 10 keywords of each topic.

The writing style also changes with the topic (Figure 7): religion has the longest sentences (28.3 words), romance the shortest sentences (17.8 words) and the shortest words (4.25 letters), and news the longest words (5.03 letters) and the most varied vocabulary (lexical diversity of 0.387). As Figure 8 shows, the keyword score classified the 10 texts in the topic of the category they come from in the Brown Corpus: the news texts had between 2.3% and 8.6% of political keywords, the religion texts about 4.2% of religious keywords and the romance texts between 2.3% and 3.3% of romance keywords.

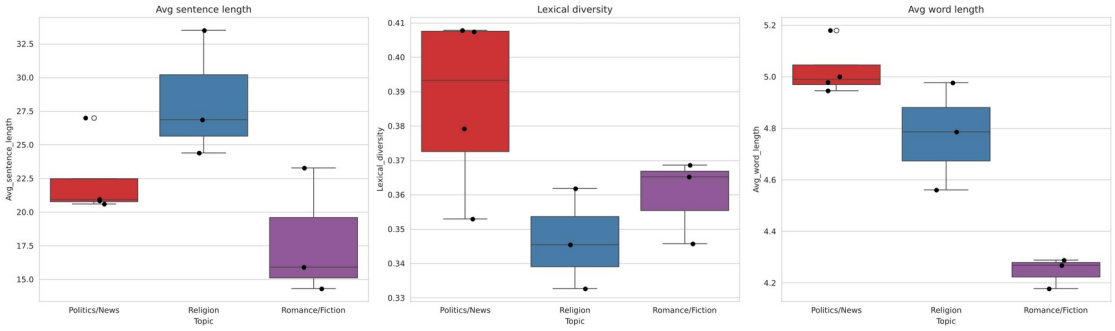


Figure 7. Writing style of each topic: sentence length, lexical diversity and word length.

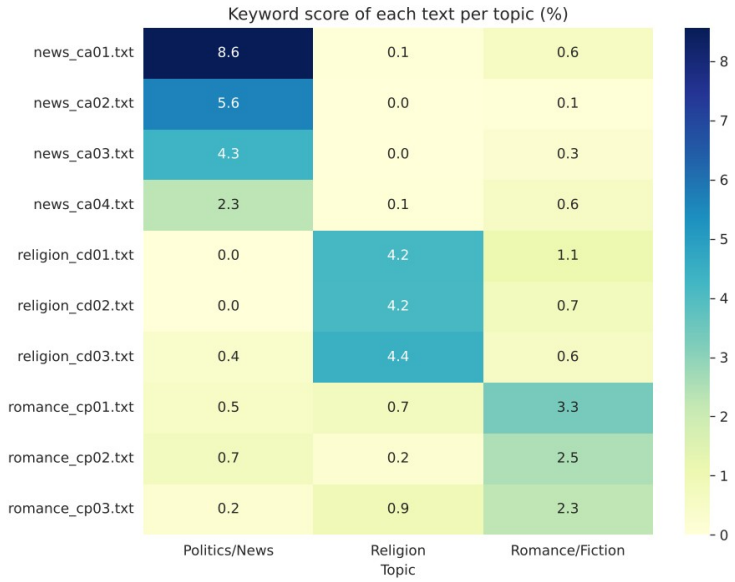


Figure 8. Keyword score of each text for each topic (%).

Table 1. Summary of the classification of the three media types.

Media	Files	Features	Result of the classification
Image	10	Hue histogram, saturation, brightness, Canny edge density	Blue 3, Grayscale 3, Orange 2, Green 1, Red 1; Smooth 3, Medium 4, Textured 3
Audio	8	Noise floor, SNR (RMS in dB), dominant frequency (FFT)	Low noise 5, Medium 1, High 2; Low pitch 2, Medium 2, High 2, No clear pitch 2
Text	10	Word frequencies (FreqDist), keywords, writing style, keyword scores	Politics/News 4, Religion 3, Romance/Fiction 3

The program also saves other charts in the Results folder, which are explained in the video: the color histogram and the Canny edges of each image, the edge density of each image, the amplitude and the FFT spectrum of the 8 sounds, the most frequent words and the number of files per class.

Link with the video explanation uploaded to drive:

https://drive.google.com/drive/folders/1ibK3pPUOVlPS5vnjybw3dWAnfyCNDlP2?usp=sharing

GitHub with the different files and results:

https://github.com/Samuel-RP22/Final-Project

Conclusions

- It was possible to integrate image, audio and text processing in one automated program: every file is detected by its extension, converted into numerical features and classified with simple rules, and the results of the three media types are summarized and plotted with the same tools.

- Simple features gave good results: the hue histogram and the Canny edges described the color and the texture of very different images, the SNR separated the constant noises from the short sounds, the FFT found pitches from 383 Hz to 8,192 Hz, and a short list of keywords was enough to find the topic of all the texts.

- Separating the processing functions from the charts made the program easier to understand, and the step by step figures show exactly how each class is obtained.

- The main limitation is that the limits of the rules were chosen by us, and a recording with some background noise, like the chirping birds, can change its class. With more data these limits could be learned with machine learning.

- This kind of program can be used to organize photo libraries by color or texture, detect noisy recordings or alarms like sirens, monitor animal and environmental sounds, and sort documents or news by topic.